

Generation Interconnection Request System Impact Study Report

For Queue Project Q16-0714

Dessau 138 kV
November 24, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0714. The Interconnection Customer proposes a 205.0 MW battery energy storage facility interconnecting at the Dessau 138 kV substation in the Virginia area. The request was received on December 14, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 344 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 34 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,171,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Cia - New Sterling 138kV line
Contingency: Cia - Cowlitz Falls 138kV line
Loading: 154.3% (Rating: 258 MVA)
DFAX: 10.0%
Pre-project loading: 146.4%
2. Facility: Papago Buttes - Cia 138kV
Contingency: Cia Substation - Papago Buttes 138kV
Loading: 153.0% (Rating: 286 MVA)
DFAX: 9.6%
Pre-project loading: 146.1%
3. Facility: Papago Buttes - Mottman 138kV
Contingency: Mottman - Papago Buttes 138kV #2
Loading: 149.4% (Rating: 321 MVA)
DFAX: 10.0%
Pre-project loading: 143.0%
4. Facility: Mottman - Papago Buttes 138kV #2
Contingency: Mottman - Papago Buttes 138kV L
Loading: 149.4% (Rating: 321 MVA)
DFAX: 10.0%
Pre-project loading: 143.0%
5. Facility: Cia - Papago Buttes 138kV Circuit 1
Contingency: Papago Buttes Substation - Cia 138kV Line 2
Loading: 145.5% (Rating: 301 MVA)
DFAX: 9.6%
Pre-project loading: 139.0%

6. Facility: Fairfax - Franconia Substation 138kV L
Contingency: Dessau - Gallows Road 138kV L
Loading: 144.5% (Rating: 205 MVA)
DFAX: 8.9%
Pre-project loading: 135.5%
7. Facility: Idylwood - Falls Church 138kV
Contingency: Centennial - Annandale 138kV
Loading: 143.8% (Rating: 200 MVA)
DFAX: 31.3%
Pre-project loading: 111.7%
8. Facility: Hemdon - Crystal Lake 138kV
Contingency: Del Amo - Grand Blanc Generating Sta Substation 138kV
Loading: 140.4% (Rating: 276 MVA)
DFAX: 15.0%
Pre-project loading: 129.2%
9. Facility: Glen Carlyn - Centennial 138kV
Contingency: Gallows Rd - Idylwood Substation 138kV
Loading: 133.2% (Rating: 202 MVA)
DFAX: 26.7%
Pre-project loading: 106.1%
10. Facility: Holmes Crk - Gallows Road 138kV L
Contingency: Gallows Road - Idylwood 138kV line
Loading: 127.7% (Rating: 219 MVA)
DFAX: 40.3%
Pre-project loading: 89.9%
11. Facility: Brookridge - Franconia 138kV
Contingency: Dessau - Gallows Rd 138kV
Loading: 126.3% (Rating: 217 MVA)
DFAX: 8.9%
Pre-project loading: 117.9%
12. Facility: Idylwood - Gallows Rd 138kV Line
Contingency: Centennial - Annandale 138kV
Loading: 125.1% (Rating: 277 MVA)
DFAX: 31.3%
Pre-project loading: 101.9%
13. Facility: Fairfax - Jeune Landing Substation 138kV
Contingency: Dessau - Gallows Rd 138kV
Loading: 124.4% (Rating: 221 MVA)
DFAX: 6.9%
Pre-project loading: 117.9%
14. Facility: Shepard - Bloomington 138kV Ckt 1
Contingency: Bloomington - Shepard 138kV Line 2
Loading: 122.4% (Rating: 313 MVA)
DFAX: 6.0%
Pre-project loading: 118.5%
15. Facility: Gallows Rd - Dessau 138kV
Contingency: Del Amo - Grand Blanc Generating Sta 138kV
Loading: 122.0% (Rating: 478 MVA)
DFAX: 53.9%
Pre-project loading: 98.9%
16. Facility: Cowlitz Falls Substation - Cia 138kV
Contingency: Cia - New Sterling 138kV Line

Loading: 121.0% (Rating: 257 MVA)

DFAX: 6.1%

Pre-project loading: 116.1%

17. Facility: Hemdon Substation - Lunenburg 138kV

Contingency: Bull Run - Del Amo 138kV

Loading: 120.7% (Rating: 248 MVA)

DFAX: 15.0%

Pre-project loading: 108.3%

18. Facility: Annandale - Centennial 138kV Line

Contingency: Idylwood - Gallows Road 138kV

Loading: 118.8% (Rating: 276 MVA)

DFAX: 26.7%

Pre-project loading: 98.9%

19. Facility: Dessau - S Houston 138kV

Contingency: Del Amo - Grand Blanc Generating Station Substation 138kV Line

Loading: 117.7% (Rating: 493 MVA)

DFAX: 46.1%

Pre-project loading: 98.6%

20. Facility: South Houston - Peavy Fall Hy 138kV

Contingency: S Houston - Fillmore 138kV

Loading: 115.1% (Rating: 261 MVA)

DFAX: 36.2%

Pre-project loading: 86.7%

21. Facility: Carderock - Swinks Mill 138kV

Contingency: Carderock - Swinks Mill 138kV

Loading: 113.8% (Rating: 367 MVA)

DFAX: 5.8%

Pre-project loading: 110.6%

22. Facility: Bloomington - Shepard 138kV #2

Contingency: Shepard - Bloomington 138kV Circuit 1

Loading: 113.5% (Rating: 337 MVA)

DFAX: 6.0%

Pre-project loading: 109.9%

23. Facility: Holmes Crk - Annandale 138kV L

Contingency: Dessau - South Houston 138kV

Loading: 111.8% (Rating: 238 MVA)

DFAX: 49.5%

Pre-project loading: 69.2%

24. Facility: Luna - Peavy Fall Hy 138kV

Contingency: Fillmore - South Houston 138kV

Loading: 110.5% (Rating: 313 MVA)

DFAX: 36.2%

Pre-project loading: 86.8%

25. Facility: Shepard - Swinks Mill 138kV

Contingency: Swinks Mill - Shepard 138kV #1

Loading: 109.4% (Rating: 372 MVA)

DFAX: 6.0%

Pre-project loading: 106.1%

26. Facility: Mottman - Bloomington 138kV Line

Contingency: Mottman - Bloomington 138kV L

Loading: 109.3% (Rating: 419 MVA)

- DFAX: 8.6%
Pre-project loading: 105.1%
27. Facility: Dulles - Lunenburg 138kV line
Contingency: Bull Run Substation - Del Amo 138kV Line
Loading: 108.4% (Rating: 246 MVA)
DFAX: 18.6%
Pre-project loading: 92.9%
28. Facility: Luna - Yavapai 138kV
Contingency: S Houston - Fillmore 138kV line
Loading: 107.8% (Rating: 353 MVA)
DFAX: 36.2%
Pre-project loading: 86.8%
29. Facility: Saranac Facility - Cowlitz Falls 138kV
Contingency: Cia - New Sterling 138kV
Loading: 107.5% (Rating: 221 MVA)
DFAX: 6.1%
Pre-project loading: 101.8%
30. Facility: Falls Church - Decatur Eergy Center Substation 138kV
Contingency: Centennial - Annanale 138kV
Loading: 105.2% (Rating: 204 MVA)
DFAX: 31.3%
Pre-project loading: 73.8%
31. Facility: New Hope - Decatur Energy Center Substation 138kV
Contingency: Dessau - S Houston 138kV
Loading: 104.1% (Rating: 219 MVA)
DFAX: 50.5%
Pre-project loading: 56.8%
32. Facility: Swinks Mill - Shepard 138kV Ckt 1
Contingency: Swinks Mill - Shepard 138kV
Loading: 103.8% (Rating: 392 MVA)
DFAX: 6.0%
Pre-project loading: 100.6%
33. Facility: Swinks Mill - Carderock 138kV Ckt 1
Contingency: Carderock - Swinks Mill 138kV
Loading: 103.4% (Rating: 404 MVA)
DFAX: 5.8%
Pre-project loading: 100.5%
34. Facility: New Hope - Mottmn 138kV line
Contingency: S Houston - Dessau 138kV
Loading: 101.0% (Rating: 203 MVA)
DFAX: 40.6%
Pre-project loading: 60.0%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cia - New Sterling 138kV L	Rebuild line with higher-capacity conductor	\$3,222,000	18% (\$593,000)
Papago Buttes - Cia 138kV	Rebuild line with higher-capacity conductor	\$1,481,000	34% (\$508,000)
Mottman - Papago Uttes 138kV Ckt 1	Rebuild line with higher-capacity conductor	\$1,528,000	18% (\$283,000)
Mottman - Papago Buttes 138kV #2	Rebuild line with higher-capacity conductor	\$964,000	18% (\$178,000)
Cia - Papago Buttes 138kV Circuit 1	Rebuild line with higher-capacity conductor	\$2,657,000	34% (\$911,000)
Fairfax - Franconia 138kV line	Rebuild line with higher-capacity conductor	\$1,855,000	100% (\$1,855,000)
Falls Churh - Idylwood 138kV	Rebuild line with higher-capacity conductor	\$1,982,000	40% (\$788,000)
Hemdon - Crystal Lake 138kV	Rebuild line with higher-capacity conductor	\$2,250,000	8% (\$182,000)
Glen Carlyn - Centennial 138kV	Rebuild line with higher-capacity conductor	\$1,662,000	28% (\$468,000)
Holmes Rk - Gallows Road 138kV	Rebuild line with higher-capacity conductor	\$2,313,000	40% (\$916,000)
Franconia - Brookridge 138kV	Rebuild line with higher-capacity conductor	\$2,632,000	100% (\$2,632,000)
Gallows Rd - Idylwood 138kV line	Rebuild line with higher-capacity conductor	\$1,514,000	40% (\$602,000)
Fairfax - Jeune Landing 138kV	Rebuild line with higher-capacity conductor	\$1,175,000	100% (\$1,175,000)
Shepard - Bloomington 138kV Ckt 1	Rebuild line with higher-capacity conductor	\$1,437,000	33% (\$476,000)
Gallows Rd - Dessau 138kV	Rebuild line with higher-capacity conductor	\$1,971,000	26% (\$514,000)
Cowlitz Falls Substation - Cia 138kV	Rebuild line with higher-capacity conductor	\$1,857,000	61% (\$1,137,000)
Hemdon Substation - Lunenburg 138kV	Rebuild line with higher-capacity conductor	\$1,188,000	100% (\$1,188,000)
Annandale - Centennial 138kV Line	Rebuild line with higher-capacity conductor	\$1,097,000	28% (\$309,000)
S Houston - Dessau 138kV	Rebuild line with higher-capacity conductor	\$2,359,000	23% (\$547,000)
South Houston - Peavy Fall Hy 138kV	Rebuild line with higher-capacity conductor	\$704,000	100% (\$704,000)
Swinks Mill - Carderock 138kV Line 2	Rebuild line with higher-capacity conductor	\$2,819,000	39% (\$1,100,000)
Shepard - Bloomington 138kV Circuit 2	Rebuild line with higher-capacity conductor	\$1,715,000	33% (\$568,000)
Holmes Creek Substation - Annandale 138kV	Rebuild line with higher-capacity conductor	\$3,860,000	99% (\$3,825,000)
Peavy Fall Hy - Luna 138kV	Rebuild line with higher-capacity conductor	\$1,778,000	100% (\$1,778,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shepard - Swinks Mill 138kV	Reconductor line	\$350,000	33% (\$116,000)
Mottman - Bloomington 138kV Line	Reconductor line	\$444,000	24% (\$104,000)
Dulles - Lunenburg 138kV line	Reconductor line	\$502,000	100% (\$502,000)
Yavapai - Luna 138kV	Reconductor line	\$708,000	100% (\$708,000)
Saranac Facility - Cowlitz Falls 138kV	Reconductor line	\$182,000	61% (\$111,000)
Falls Church - Decatur Eergy Center Substation 138kV	Reconductor line	\$447,000	40% (\$178,000)
Decatur Energy Center - New Hope 138kV L	Reconductor line	\$218,000	99% (\$216,000)
Swinks Mill - Shepard 138kV Ckt 1	Reconductor line	\$218,000	33% (\$72,000)
Sinks Mill - Carderock 138kV	Reconductor line	\$970,000	39% (\$378,000)
New Hope - Mottmn 138kV line	Reconductor line	\$887,000	62% (\$549,000)
	Total allocated to project		\$26,171,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.